

Floris van Doorn

vdoorn@math.uni-bonn.de

<https://florisvandoorn.com>

 0000-0003-2899-8565

 fpvandoorn

Google Scholar: 1.9k citations, h-index 13.

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Education and Employment

- 2023 – present W2 professor, Mathematical Institute, University of Bonn.
- 2021 – 2023 Postdoctoral Associate, Mathematics Department, University of Paris-Saclay.
- 2018 – 2021 Postdoctoral Associate, Mathematics Department, University of Pittsburgh.
- 2013 – 2018 Ph.D. in Pure and Applied Logic, Carnegie Mellon University.
- Dissertation: *On the Formalization of Higher Inductive Types and Synthetic Homotopy Theory*.
- Advisors: Prof. Jeremy Avigad, Prof. Steve Awodey.
- 2011 – 2013 M.Sc. (cum laude), Mathematical Sciences, Utrecht University.
- Thesis: *Explicit convertibility proofs in Pure Type Systems*.
- Advisor: Prof. Freek Wiedijk.
- 2008 – 2011 B.Sc. (cum laude), Mathematics, Utrecht University.
- 2008 – 2011 B.Sc. (cum laude), Physics and Astronomy, Utrecht University.

Publications

- 2024 *Integrals Within Integrals: A Formalization of the Gagliardo-Nirenberg-Sobolev Inequality*, Floris van Doorn, Heather Macbeth. Interactive Theorem Proving (ITP) 2024.
- 2023 *Formalising the h-principle and sphere eversion*, Patrick Massot, Floris van Doorn, Oliver Nash. Certified Programs and Proofs (CPP) 2023.
- 2021 *Progress on a Perimeter Surveillance Problem*, Jeremy Avigad, Floris van Doorn. Thirty-Fifth Annual International Conference on Autonomous Systems (ICAS).
- 2021 *Formalized Haar Measure*, Floris van Doorn. Interactive Theorem Proving (ITP).
- 2020 *Maintaining a Library of Formal Mathematics*, Floris van Doorn, Gabriel Ebner, and Robert Y. Lewis. 13th Conference on Intelligent Computer Mathematics (CICM).
- 2020 *Sequential Colimits in Homotopy Type Theory*, Kristina Sojakova, Floris van Doorn, Egbert Rijke. Thirty-Fifth Annual ACM/IEEE Symposium on Logic in Computer Science (LICS).
- 2020 *A Formal Proof of the Independence of the Continuum Hypothesis*, Jesse Michael Han and Floris van Doorn. Certified Programs and Proofs (CPP).
- 2020 *The Lean Mathematical Library*, the mathlib community.¹ Certified Programs and Proofs (CPP).
- 2019 *A Formalization of Forcing and the Unprovability of the Continuum Hypothesis*, Jesse Michael Han and Floris van Doorn. Interactive Theorem Proving (ITP).
- 2018 *Higher Groups in Homotopy Type Theory*, Ulrik Buchholtz, Floris van Doorn, Egbert Rijke. Logic in Computer Science (LICS).

¹This was a paper written collectively by the contributors to `mathlib`. I wrote part of the paper.

- 2017 *Homotopy Type Theory in Lean*, Floris van Doorn, Jakob von Raumer, Ulrik Buchholtz. 8th International Conference on Interactive Theorem Proving (ITP).
- 2016 *Constructing the Propositional Truncation using Non-recursive HITs*, Floris van Doorn. The 5th ACM SIGPLAN Conference on Certified Programs and Proofs (CPP).
- 2015 *The Lean Theorem Prover (System Description)*, Leonardo de Moura, Soonho Kong, Jeremy Avigad, Floris van Doorn, Jakob von Raumer. The 25th jubilee edition of the International Conference on Automated Deduction (CADE). This paper won the [Skolem Award](#) in 2025.
- 2014 *The Structural Theory of Pure Type Systems*, Cody Roux and Floris van Doorn. LNCS Advanced Research in Computing and Software Science.
- 2013 *Explicit Convertibility Proofs in Pure Type Systems*, Floris van Doorn, Herman Geuvers, Freek Wiedijk. Workshop on Logical Frameworks and Meta-languages: Theory and Practice (LFMTP).

Grants and Awards

- 2026 HCM Grant: Editor for the Annals of Formalized Mathematics (7k €)
- 2025 ERC Synergy grant *HALF: Harmonic Analysis and Lean Formalization* 2026–2032 with Christoph Thiele (6.4M €).
- 2025 [Skolem award](#) for the paper *The Lean Theorem Prover (System Description)*.
- 2025 Principal Investigator for the renewal of the DFG Excellence Strategy – EXC-2047/1 – 390685813 (24 co-PIs, 45.3M €).
- 2024 Application partner of NFDI 29/1 [MaRDI](#) (Mathematische Forschungsdateninitiative).
- 2021 Labex Mathématiques Hadamard 2-year Postdoctoral Fellowship in the area Mathematics for Artificial Intelligence.
- 2009 Royal Holland Society of Sciences and Humanities “Young Talent Award” in mathematics.

Preprints and Unpublished Work

- 2026 *Formalizing Carleson’s Theorem in Lean*, Floris van Doorn et al. (14 authors). Preprint.
- 2025 *The Equational Theories Project: Advancing Collaborative Mathematical Research at Scale*, Terence Tao et al. (I am one of the 34 authors). Preprint.
- 2025 *Carleson operators on doubling metric measure spaces*, Lars Becker, Floris van Doorn, Asgar Jamneshan, Rajula Srivastava, Christoph Thiele. Preprint.
- 2024 *A blueprint for the formalization of Carleson’s theorem on convergence of Fourier series*, Lars Becker, María Inés de Frutos-Fernández, Leo Diederich, Floris van Doorn, Sébastien Gouëzel, Asgar Jamneshan, Evgenia Karunus, Edward van de Meent, Pietro Monticone, Jasper Mulder-Sohn, Jim Portegies, Joris Roos, Michael Rothgang, Rajula Srivastava, James Sundstrom, Jeremy Tan, Christoph Thiele. Blueprint for a formalization project.
- 2022 *Designing a general library for convolutions*, Floris van Doorn. Preprint.
- 2016 *Logic and Proof*, Jeremy Avigad, Robert Y. Lewis, Floris van Doorn. Online textbook for an introductory course to logic and proof assistants.
- 2014 *Propositional Calculus in Coq*, Floris van Doorn. Short article on arXiv.

Given Talks

- 2026 *The Carleson Project: Collaboration using Formalization*, Lean for the Curious Mathematician, Cortona.
- 2026 *Towards formalizing variation estimates for commuting transformations*, ICERM workshop “Techniques and Tools for the Formalization of Analysis”, Providence.
- 2026 *Towards formalizing variation estimates for commuting transformations*, seminar “Formalisation of mathematics with interactive theorem provers”, Cambridge.
- 2026 *Lean: Collaboration using Formalization*, Institute-wide colloquium talk at the Institute of Science and Technology Austria.
- 2026 *Lean: Collaboration using Formalization*, Colloquium talk at SMS Spring Meeting: Formalization and Proof Assistants.
- 2026 *Lessons from the Carleson project*, Lean Together, online
- 2025 *Lean: Collaboration using Formalization*, MaRDI Annual Meeting 2025, Munich.
- 2025 *The Carleson project: a collaborative formalization* (joint with María Inés de Frutos Fernández). MATRIX-MFO Tandem Workshop: Machine Learning and AI for Mathematics, Oberwolfach.
- 2025 *The Carleson Project: Formalization and Collaboration*. Big proof: formalizing mathematics at scale, Isaac Newton Institute, Cambridge.
- 2025 *The Carleson Project: Collaboration using Formalization*. Simons Institute for the Theory of Computing and SLMath Joint Workshop: AI for Mathematics and Theoretical Computer Science.
- 2025 *Progress report on the Carleson Project*, Lean together, online.
- 2025 *Formalizing modern mathematics in Lean*, LEAN meets MaRDI and OSCAR, Berlin.
- 2024 *Formalizing a proof of Carleson’s theorem*, Homotopy Type Theory Electronic Seminar Talks (HoTTTEST), online.
- 2024 *Carleson operators on doubling metric measure spaces* (joint with Christoph Thiele). Special Topic School: Maximal Operators and Applications, Bonn.
- 2024 *Towards a formalized proof of Carleson’s theorem*, Hausdorff trimester.
- 2024 *The Sobolev inequality in Lean*, Informal Formalization Seminar.
- 2024 *The internals of Lean*, 48 hours in Rome.
- 2023 *Mathematics in Lean*, Hausdorff School, Bonn.
- 2023 *The Independence of the Continuum Hypothesis in Lean*, LFTCM, Düsseldorf.
- 2023 *Formalizing sphere eversion using Lean’s mathematical library*, CALCO 2023 & MFPS XXXIX, Bloomington, Indiana.
- 2023 *Tutorial on interactive theorem proving in Lean*, Logic Colloquium, Milan.
- 2023 *What can we learn from formalizations in homotopy type theory?*, Formalization of Cohomology Theories, Banff International Research Station.
- 2023 *Formalizing sphere eversion in Lean*, INRIA, Nantes.
- 2022 *Formalizing sphere eversion in Lean*, IRIF, Paris.
- 2022 *Lessons Learned from Formalizing Local Convex Integration*, Lean in Lyon.
- 2021 *Automating Concept Equivalence in Dependent Type Theory*, 6th Conference on Artificial Intelligence and Theorem Proving (AITP).
- 2020 *Tactics in Lean*, HCM Workshop: Mathematical Language and Practical Type Theory, Bonn.

- 2019 *A Formal Abstract of the Classification of Finite Simple Groups*, Vietnam – USA Joint Mathematical meeting.
- 2018 *Towards Spectral Sequences for Homology*, Homotopy Type Theory Electronic Seminar Talks (HoTTEST), online.
- 2018 *Spectral Sequences in Homotopy Type Theory*, Types, Homotopy Type theory, and Verification, Hausdorff Research Institute for Mathematics, Bonn.
- 2017 *Homotopy Type Theory in Lean*, Computer-aided mathematical proof, Cambridge.
- 2017 *The Lean HoTT Library*, Big Proof, Isaac Newton Institute, Cambridge.
- 2017 *Eilenberg–MacLane Spaces in Homotopy Type Theory*, ASL North American annual meeting, Boise.
- 2016 *Homotopy Type Theory in Lean*, Univalent Foundations and Proof Assistants, International Congress on Mathematical Software (ICMS), Berlin.
- 2016 *Homotopy Type Theory in Lean*, Workshop on Homotopy Type Theory / Univalent Foundations, Porto.
- 2016 *Reducing Higher Inductive Types to Quotients*, Workshop on Homotopy Type Theory and Univalent Foundations of Mathematics (HoTT/UF), Fields Institute Toronto.
- 2016 *The Lean Theorem Prover and Homotopy Type Theory*, joint with Jeremy Avigad. Workshop on Homotopy Type Theory and Univalent Foundations of Mathematics (HoTT/UF), Fields Institute Toronto.

Teaching

- 2026 Instructor for Formalized Mathematics in Lean (Lecture course, Bonn).
- 2025 Instructor for Formalized Mathematics in Lean (Lecture course, Bonn).
- 2025 Instructor for The Logic of Proof Assistants (Lecture course, Bonn).
- 2024 Instructor for Formalized Mathematics in Lean (Practical training course, Bonn).
- 2024 Instructor for Collaborative Formalization Project in Analysis (Graduate Seminar, Bonn).
- 2023 Instructor for Formalized Mathematics in Lean (Practical training course, Bonn).
- 2021 Instructor for Abstract Algebra (Pitt).
- 2020 Instructor for Topics in Geometry (Pitt).
- 2019 Instructor for Calculus I (Pitt).
- 2016 TA for Differential and Integral Calculus with Russell C. Walker (CMU).
- 2015 TA for Logic and Mathematical Inquiry with Jeremy Avigad (CMU).
- 2015 TA for Game Theory with Adam Bjorndahl (CMU).
- 2014 TA for Formal Logic with Steve Awodey (CMU).
- 2012 TA for Discrete Mathematics with Han Hoogeveen (Utrecht).
- 2011 TA for Foundations of Mathematics with Jaap van Oosten (Utrecht).

Supervision

Postdocs

- Leopold Mayer (Oct 26–ongoing)
- Michael Rothgang (Oct 24–ongoing)

- María Inés de Frutos Fernández (Oct 24–Oct 26) → CNRS in Paris-Cité

PhD students

- Koen Bresters (Nov 26–ongoing)
- Lua Viana Reis (Apr 26–ongoing)
- Arend Mellendijk (Oct 24–ongoing)

Including ongoing projects, I have supervised 4 Master thesis projects and co-supervised 8 more. I have also supervised 7 Bachelor thesis projects, and co-supervised 3 more. I have mentored 5 student research assistants.

Every semester, starting in the summer semester of 2024, I have organized the seminar for the Formalized Mathematics group in Bonn. This seminar features student presentations from their Bachelor or Master projects and presentations by members of the group or invited guests.

More details and the research output of my group can be found on the webpage <https://florisvandoorn.com/formalized-mathematics>.

Service

2023 – present	Leader for the interdisciplinary research unit (IRU) on Formalized Mathematics in Bonn.
2019 – present	Maintainer for Lean’s mathematical library Mathlib . ²
2026	Program Committee for Interactive Theorem Proving (ITP).
2025	Organizer for Simons Institute for the Theory of Computing and SLMATH Joint Workshop: AI for Mathematics and Theoretical Computer Science.
2024	Program Committee for Interactive Theorem Proving (ITP).
2024	Scientific organizer for the Workshop Computer-verified proofs: 48 hours in Rome.
2020 – 2022	Program Committee for Certified Programs and Proofs (CPP).
2021	Program Committee for the Workshop on Homotopy Type Theory / Univalent Foundations (HoTT/UF) 2021.
2021	Problem Committee for the Proof Ground 2021 Interactive Proving Contest.

Besides PC memberships, I have also reviewed 13 manuscripts as external reviewer for the following conferences and journals:

Conference on Intelligent Computer Mathematics (CICM), Experimental Mathematics Foundations of Software Science and Computation Structures (FOSSACS), Interactive Theorem Proving (ITP), Journal of Automated Reasoning, Logical Methods in Computer Science (LMCS), Mathematical Structures in Computer Science, Transactions on Computational Logic, Types Conference and Workshop on Homotopy Type Theory / Univalent Foundations (HoTT/UF). Furthermore, I have been an anonymous reviewer for 12 grant applications, an examiner for the 1 PhD thesis defense, and I have provided one anonymous external PhD thesis evaluation.

²I am one of the 28 maintainers of Lean’s mathematical library, which involves making high-level decisions about the organization and direction of the library, reviewing and approving new contributions to the library, representing the Lean community, helping new Lean users, giving tutorials at “Lean for the Curious Mathematician” meetings and coordinating activities (e.g. workshops) related to Lean. The Mathlib project was awarded the [2026 Demailly Prize](#)

Outreach / Extracurricular Service

- 2024 – present Organizer of a weekly 2-hour “Lean hacking session” extracurricular meeting for students in Bonn. During this meeting students work on Lean projects, can discuss Lean topics and ask questions. Typical attendance has grown from ~ 5 to $\sim 10 - 15$ students.
- 2026 Interviewee for the article *Mathematicians and AI in behind-the-scenes battle over what’s true*, New Scientist, 2 October 2026.
- 2025 Interviewee for the article *Was passiert, wenn niemand mehr die Mathematik versteht?*, Spektrum der Wissenschaft 33/2025.
- 2025 Public lecture *Primes, Proofs and Computers*, Bonn Math Night.
- 2020 Interviewee for the podcast *De fascinatievoller deel 16* (Dutch *The fascination follower part 16*).
- 2009 – 2013 Trainer of the Dutch Mathematical Olympiad.
- 2008 – 2013 Volunteer for the “Vierkant voor Wiskunde” mathematics summer camps.
- 2012 – 2013 Chairman of the Benelux Mathematical Olympiad 2013.
- 2011 – 2012 Treasurer of the Dutch University Mathematical Olympiad 2012.
- 2009 – 2011 IT committee member for the International Mathematical Olympiad 2011.
- 2009 – 2010 Head awards ceremony of the Benelux Mathematical Olympiad 2010.
- 2008 – 2009 Secretary of the Benelux Mathematical Olympiad 2009.

I have written the following articles in mathematics magazines.

- 2012 *Roosters Kleuren* (Dutch, *coloring grids*), Floris van Doorn. Vakidoot 11/12(5):20-21
- 2011 *Op weg naar IMO 2011. IMO 2008, Opgave 5* (Dutch, *Towards IMO 2011. IMO 2008, Problem 5*), Floris van Doorn. Euclides 86(4):142-143 (Feb 2011)
- 2008 *Wiskunde in Vietnam* (Dutch, *Mathematics in Vietnam*), Floris van Doorn. Pythagoras 47(3):20 (jan. 2008).
- 2008 *De eerste ronde van de Nederlandse Wiskunde Olympiade 2006 – 2008. Een bundel met opgaven en uitgebreide uitwerkingen* (Dutch, *The first round of the Dutch Mathematical Olympiad 2006 – 2008. A booklet with problems and extensive solutions*), Floris van Doorn, Alexander van Hoorn, Maarten Roelofsma. Unpublished booklet.

International Mathematics Competitions

- 2012 First prize at the International Mathematics Competition for University Students.
- 2011 Second prize at the International Mathematics Competition for University Students.
- 2010 Second prize at the International Mathematics Competition for University Students.
- 2008 Silver medal at the International Mathematical Olympiad.

Languages

Dutch (native), English (fluent), German (level B2–C1), French (intermediate).

Computer languages: Fluent in Lean, $\text{\LaTeX}$, Mathematica, Coq.
Experience with Python, C, Standard ML.